

Complex Analysis – 3

Принцип аргумента, теорема Руше и подсчёт нулей

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Теоретический минимум.

Опр. For a closed curve γ not passing through a , the *winding number* is $n(\gamma, a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z-a} \in \mathbb{Z}$; it counts how many times γ goes around a .

Теорема 1 (принцип аргумента) *If f is meromorphic inside and on a simple closed contour γ , with no zeros or poles on γ , then*

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{f'(z)}{f(z)} dz = Z - P,$$

where Z and P are the numbers of zeros and poles inside γ , counted with multiplicity; equivalently $Z - P$ is the winding number of $f \circ \gamma$ around the origin.

Теорема 2 (Rouché) *If $|f(z) - g(z)| < |g(z)|$ on γ , then f and g have the same number of zeros inside γ .*

Теорема 3 (Hurwitz) *If $f_n \rightarrow f$ locally uniformly, all f_n are non-vanishing on a domain and $f \not\equiv 0$, then f is non-vanishing; consequently a locally uniform limit of univalent functions is univalent or constant.*

Факт 1 Открытость: a non-constant holomorphic map is open, and this is exactly the argument principle read backwards; the maximum modulus principle is its immediate corollary.

1. Prove the argument principle and deduce Rouché's theorem from it. Then give the two-line proof of the fundamental theorem of algebra, and count the roots of $z^4 - 5z + 1$ in $|z| < 1$ and in $1 < |z| < 2$.

2. (S.T. Yau CSMC 2018) Let $U \subset \mathbb{C}$ be an open set containing the closed unit disc $\overline{\mathbb{D}}$ and let $f: U \rightarrow \mathbb{C}$ be a non-constant holomorphic function with $|f(z)| = 1$ for all $z \in \partial\mathbb{D}$. Prove that $\mathbb{D} \subseteq f(\mathbb{D})$.
3. (S.T. Yau CSMC 2020) Let $\Omega \subseteq \mathbb{C}$ be a domain containing the closed unit disc and let $f: \Omega \rightarrow \mathbb{C}$ be holomorphic with $f(0) = 0$ and $|f(e^{i\theta})| \geq 3$ for all θ . If $\lambda_1, \dots, \lambda_N$ are all the zeros of $1 - f(z)$ in $\mathbb{D}$, counted with multiplicity, prove that $|\lambda_1 \cdots \lambda_N| < \frac{1}{2}$.
4. Prove Hurwitz's theorem, and deduce that the limit of a locally uniformly convergent sequence of univalent functions is univalent or constant. Give an example showing that «or constant» cannot be dropped.
5. Prove that a holomorphic function f on a neighbourhood of $\overline{\mathbb{D}}$ with $|f(z)| < 1$ on $|z| = 1$ has exactly one fixed point in $\mathbb{D}$, and that the iterates of f converge to it.